High energy, high luminosity *muon-hadron* collider at CERN

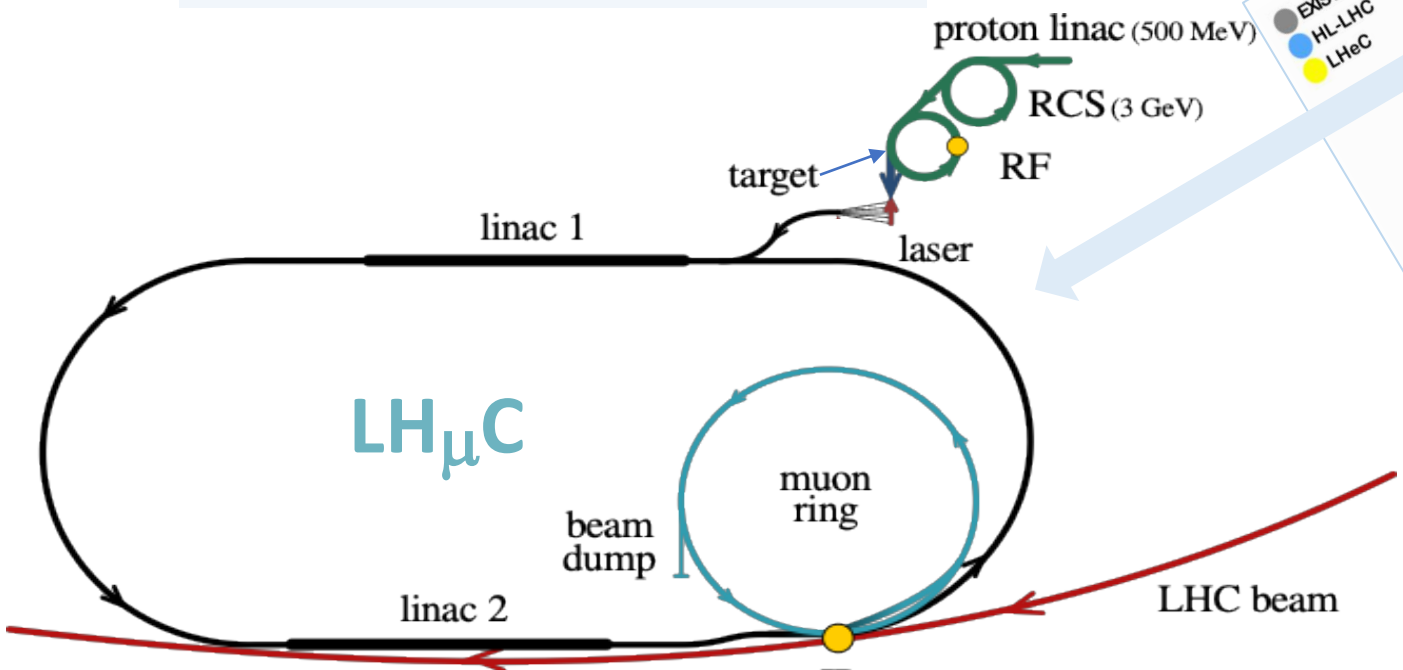
Krzysztof PIOTRZKOWSKI

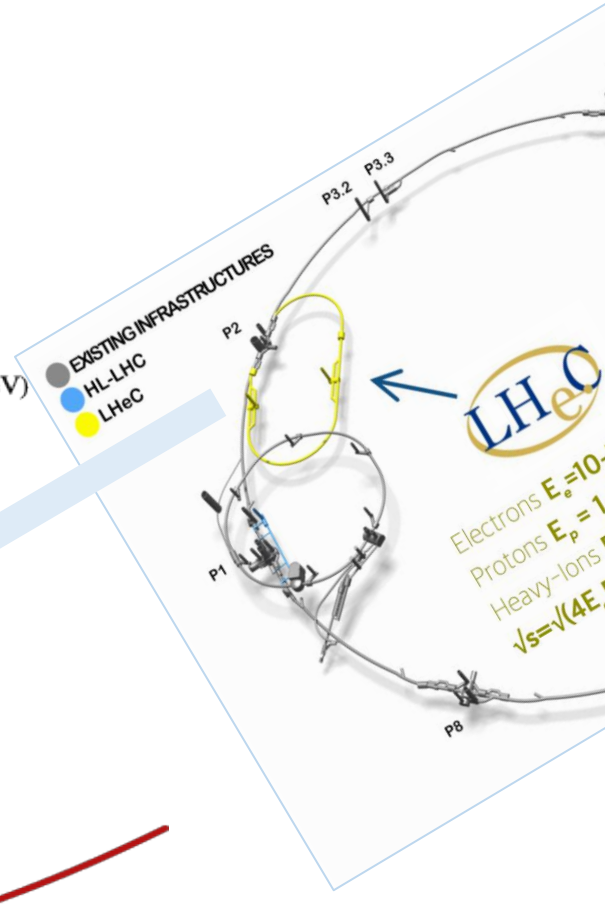
AGH University of Krakow

AGH

on behalf of HELP4CERN Study Group

DIS 2026





When it all began…

PTEP

Prog. Theor. Exp. Phys. **2022** 053B02(16 pages)
DOI: 10.1093/ptep/ptac059

μTRISTAN

Yu Hamada[1], Ryuichiro Kitano[1,2], Ryutaro Matsudo[1], Hiromasa Takaura[1,⋆], and Mitsuhiro Yoshida[2,3]

[1] *KEK Theory Center, Tsukuba 305-0801, Japan*
[2] *Graduate University for Advanced Studies (Sokendai), Tsukuba 305-0801, Japan*
[3] *KEK Accelerator Department, Tsukuba 305-0801, Japan*
⋆E-mail: takaura.phys@gmail.com

Received January 21, 2022; Revised March 18, 2022; Accepted March 28, 2022; Published March 30, 2022

The ultra-cold muon technology developed for the muon $g-2$ experiment at J-PARC provides a low-emittance μ^+ beam which can be accelerated and used for realistic collider experiments. We consider the possibility of new collider experiments by accelerating the μ^+ beam up to 1 TeV. Allowing the μ^+ beam to collide with a high-intensity e^- beam at the TRISTAN energy, $E_{e^-} = 30$ GeV, in a storage ring with the same size as TRISTAN (a circumference of 3 km), one can realize a collider experiment with the center-of-mass energy $\sqrt{s} = 346$ GeV, which allows the production of Higgs bosons through vector boson fusion processes. We estimate the deliverable luminosity with existing accelerator technologies to be at the level of $5 \times 10^{33}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$, with which the collider can be a good Higgs boson factory. $\mu^+\mu^+$ colliders up to $\sqrt{s} = 2$ TeV are also possible using the same storage ring. They have the capability of producing the superpartner of the muon up to TeV masses.

PTEP

Prog. Theor. Exp. Phys. **2026** 033C04 (20 pages)
DOI: 10.1093/ptep/ptag031

μLHC: Antimuon Ring and High Luminosity-LHC-Based μ^+p Collider

D. Akturk[1], A. C. Canbay[2], H. Dagistanli[3], B. Dagli[1], U. Kaya[4], B. Ketenoglu[5], A. Kilic[6], F. Kocak[6], A. Ozturk[7], S. Sultansoy[1,8], I. Tapan[6], and F. Zimmermann[9]

[1] *Department of Materials Science and Nanotechnology Engineering, TOBB University of Economics and Technology, Ankara, Türkiye*
[2] *Department of Physics, Ankara University, Ankara, Türkiye*
[3] *Private Radikal Educational Institutions, Izmir, Türkiye*
[4] *Institute of Accelerator Technologies, Ankara University, Ankara, Türkiye*
[5] *Department of Engineering Physics, Ankara University, Ankara, Türkiye*
[6] *Department of Physics, Bursa Uludag University, Bursa, Türkiye*
[7] *Turkish Accelerator and Radiation Laboratory (TARLA), Ankara, Türkiye*
[8] *Institute of Physics, Ministry of Science and Education, Baku, Azerbaijan*
[9] *European Organization for Nuclear Research (CERN), Geneva, Switzerland*

Received August 5, 2025; Revised January 14, 2026; Accepted January 24, 2026; Published March 2, 2026

A conceptual design and a performance evaluation of the High Luminosity-Large Hadron Collider (HL-LHC)-based antimuon–proton collider μLHC are presented. Leveraging the μTRISTAN concept based on established Japan Proton Accelerator Research Complex (J-PARC) ultra-cold μ^+ beam technology, μLHC will give the opportunity to achieve a 5.3-TeV center-of-mass energy, significantly surpassing the Electron Ion Collider (EIC) and Large Hadron-electron Collider (LHeC). Two booster ring options for μ^+ acceleration, namely, a μTRISTAN-based and a repurposed LHeC Energy Recovery Linac (ERL)-based system, are explored. Achievable luminosities are predicted to exceed $10^{33}\mathrm{cm}^{-2}\mathrm{s}^{-1}$. The μLHC offers substantially wider kinematic plane coverage, particularly in small-x and high-Q^2 regions, significantly contributing to QCD basics and Higgs boson properties. Its unique potential for Beyond the Standard Model physics extends to muon-related phenomena like excited muons, color-octet muons, leptoquarks, and contact interactions. A possible detector concept is also outlined. Given the maturity of ultra-cold μ^+ beam technology, μLHC is highly feasible for earlier realization than the muon collider, positioning it as a critical tool for the future of high-energy physics.

Bringing muons to HL-LHC

LHμC: Step 1

Following/adapting μTRISTAN proposal…

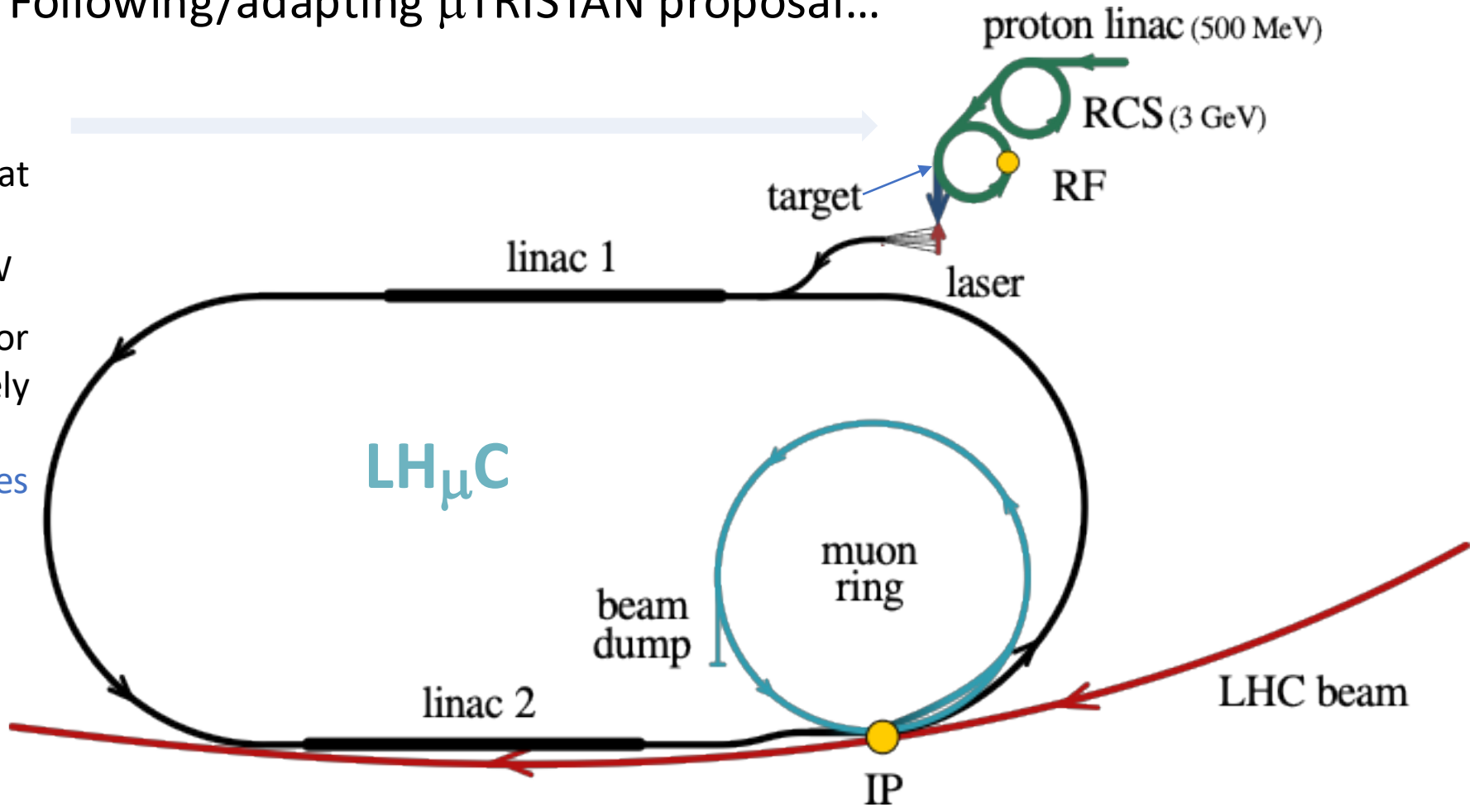
Pion production about **x 100**

Assume 3 GeV RCS similar to that at J-PARC: 8.10^{13} proton bunch
but 25 → 50 Hz, 1 → 2 MW

Use another ring with RF cavity for energy compensation to repetitively hit 1 cm graphite target
→ produce 40 pion pulses separated by **300 ns**

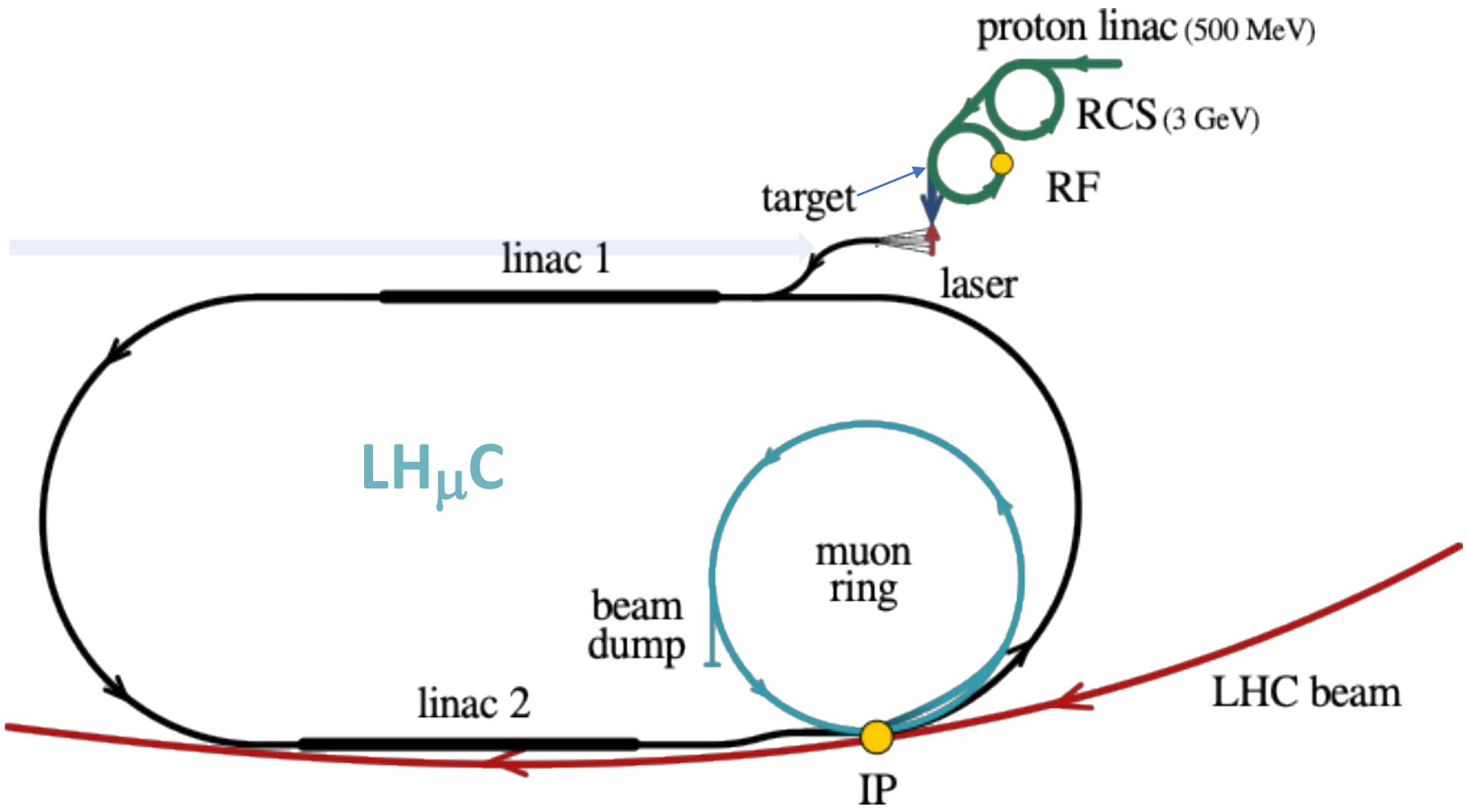
Target heat load is about 140 kW

Correction due to



LHμC: Step 2

By use of sophisticated multi-layered scheme, combined, pion stopping, surface muon production and cooling efficiency increase flux of ultra-cold muons by about **10^6**



Ultra-cold μ^+ beams at J-PARC

PHYSICAL REVIEW LETTERS **134,** 245001 (2025)

Editors' Suggestion | Featured in Physics

Acceleration of Positive Muons by a Radio-Frequency Cavity

Acceleration of positive muons from thermal energy to 100 keV has been demonstrated. Thermal muons were generated by resonant multiphoton ionization of muonium atoms emitted from a sheet of laser-ablated aerogel. The thermal muons were first electrostatically accelerated to 5.7 keV, followed by further acceleration to 100 keV using a radio-frequency quadrupole with an intensity of 2×10^{-3} μ^+/pulse. The transverse normalized rms emittance of the accelerated muons in the horizontal and vertical planes were $0.85 \pm 0.25(\text{stat})^{+0.22}_{-0.13}(\text{syst})$ π mm mrad and $0.32 \pm 0.03(\text{stat})^{+0.05}_{-0.02}(\text{syst})$ π mm mrad, respectively. The measured emittance values demonstrated phase-space reduction by a factor of 2.0×10^2 (horizontal) and 4.1×10^2 (vertical) allowing good acceleration efficiency. These results pave the way to realize the first-ever muon accelerator for a variety of applications in particle physics, material science, and other fields.

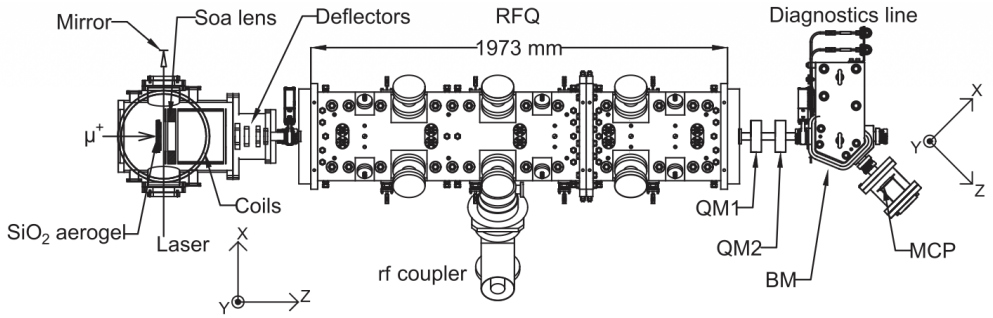
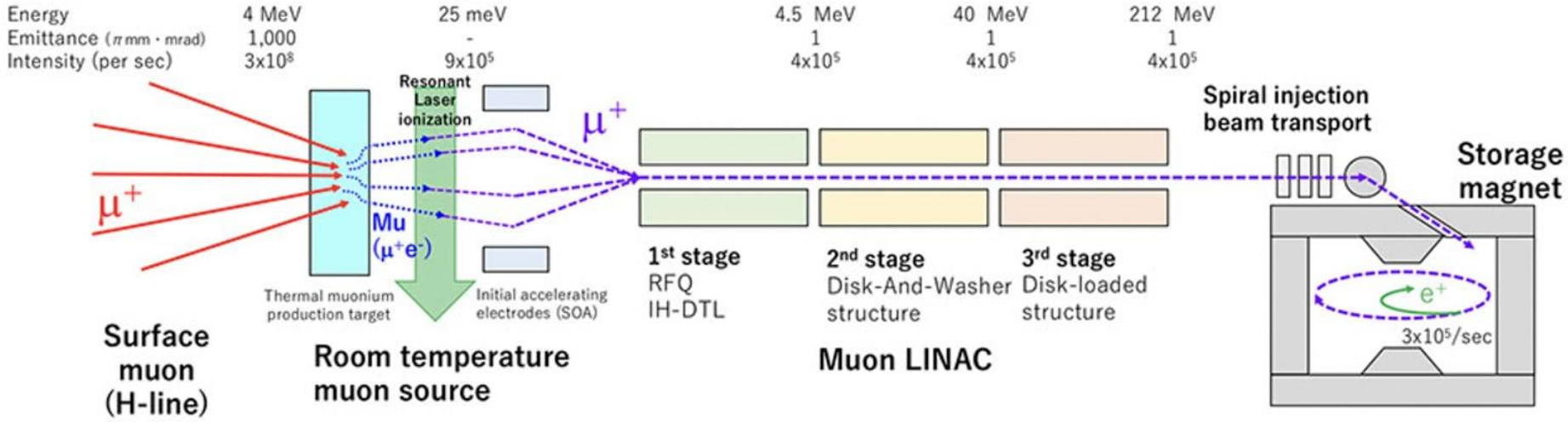


FIG. 1. Top view of the experimental setup. The surface muon beam from the MUSE S line is stopped inside a SiO_2 aerogel target. The muonium atoms emitted from the target are ionized by an ionization laser to produce ultraslow muons. The laser travels horizontally and at a 2 mm distance from the target and is reflected by a mirror for the Doppler-free excitation. The ultraslow muons are transported by the Soa lens at 5.7 keV and accelerated to 100 keV by an RFQ. Muons passing through a diagnostic line are detected by an MCP.



g-2/EDM Experiment

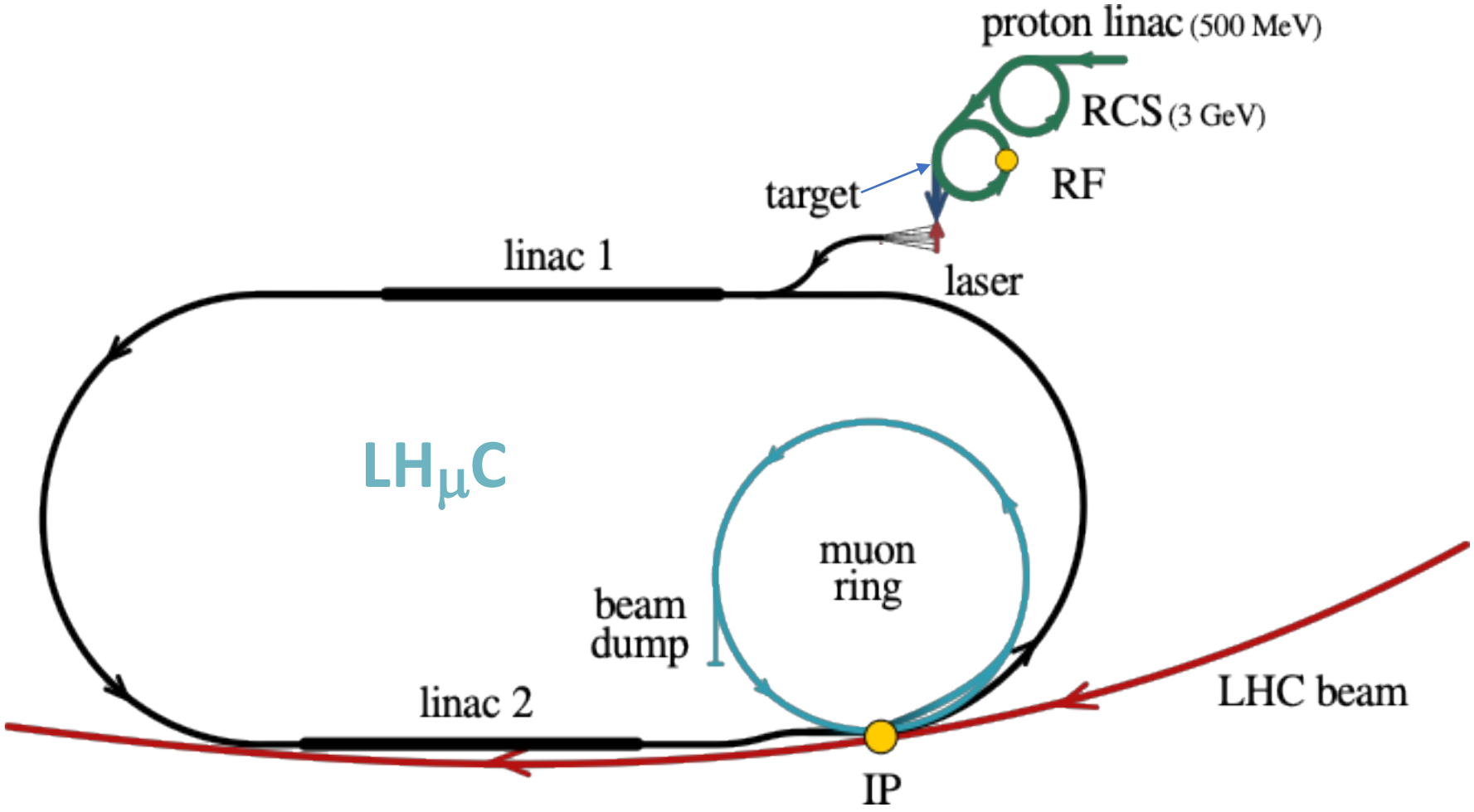
LHμC: Step 3

Muon booster

Injection energy = 0.5 GeV by muon pre-acceleration.

By **re-use of ERL linacs** one gets: first, 84.5 GeV muons by **re-using LHeC arcs**, and then up to 1 TeV by using new vFFA arcs (without RF cavities). In total it requires **42 turns** for booster circumference of 8.8 km.

To get 2 TeV beam, upgraded linacs of 15 GeV are considered.



???

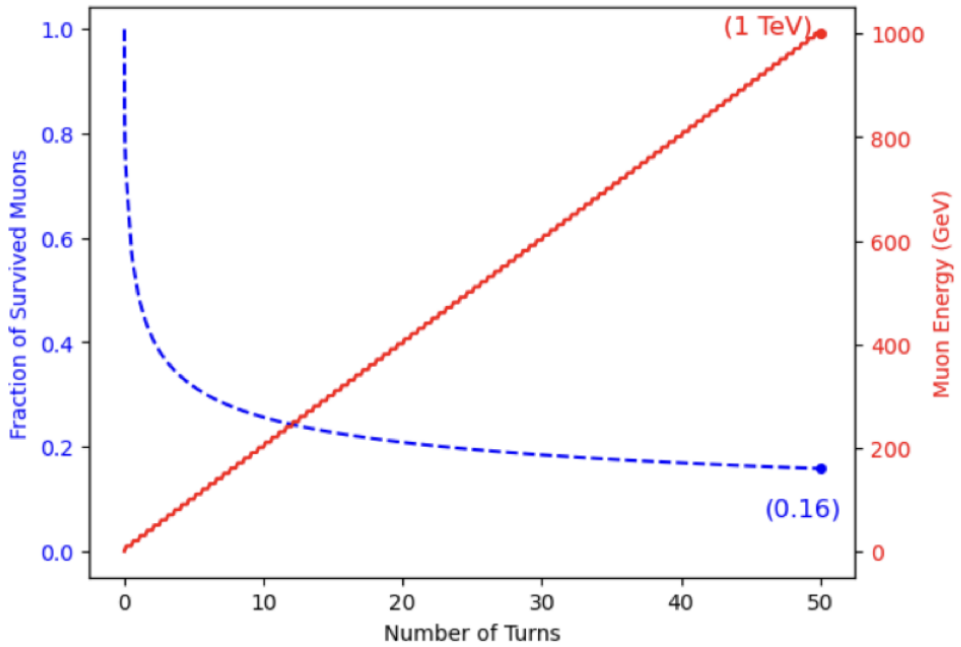
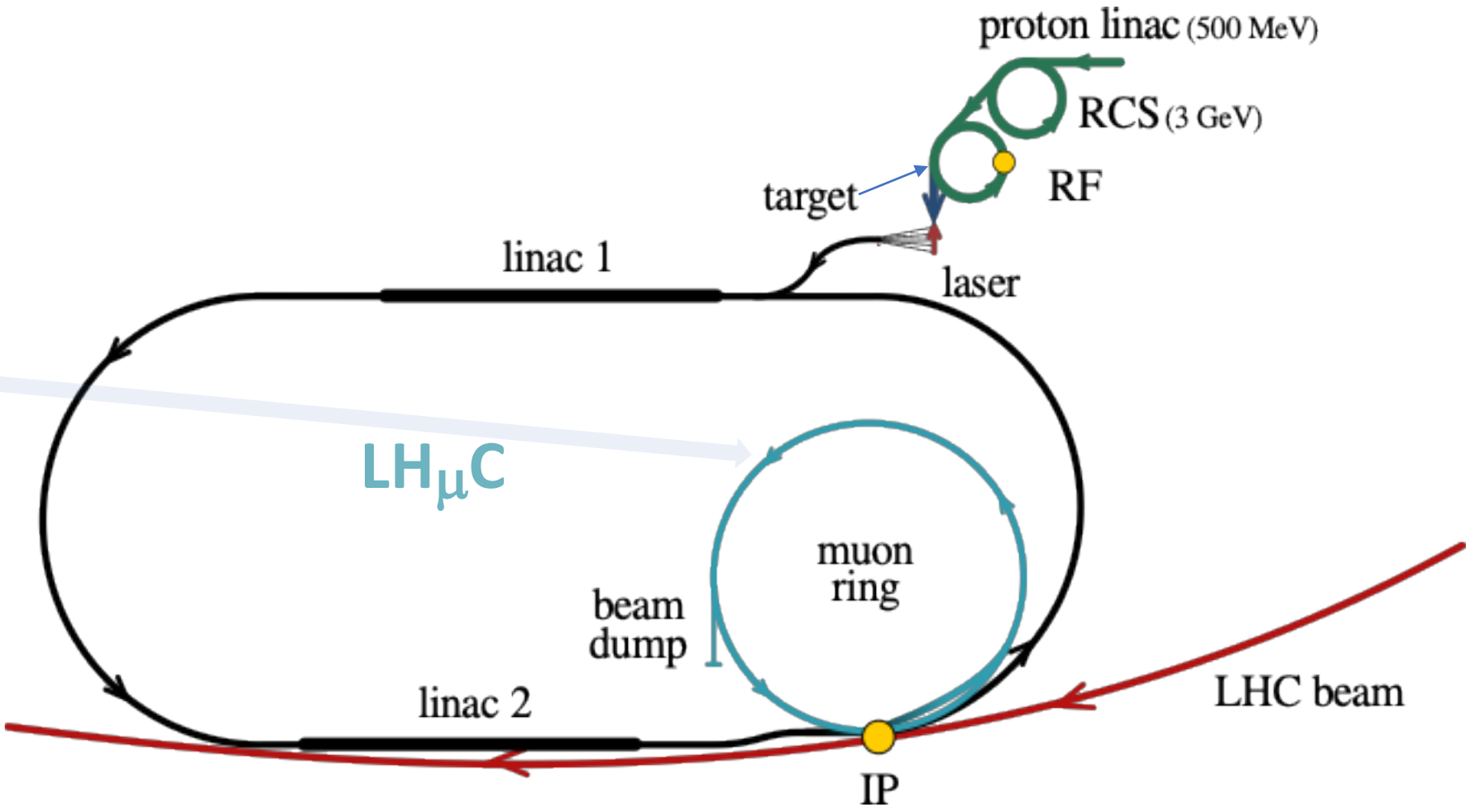


Figure 7. Muon beam evolution in the ERL-based booster Ring: energy gain and fraction of survived muons per turn.

LHμC: Step 4

Main muon ring of 3.6 km circumference is optimal for 1 TeV top energy

Ring of 6.6 km is needed to handle 2 TeV muons to maximize potential, but extra civil work similar!



LHμC: Performance

Bunch spacing is 300 ns, hence there is possibility of proton bunches twice as populous and very small crossing angle is sufficient to avoid parasitic collisions outside IP

Proton beam emittance "fixes" minimal **beam sizes at IP of 4.3 μm**, assuming HL-LHC optics and stand-alone running (no collisions at IP1 and 5)

Assuming* bunches entering booster contain $4.5\,10^{10}$ muons, peak luminosity of **above 10^{34} cm^{-2}s^{-1}** is expected for 1 TeV, and possibly **5.10^{34} cm^{-2}s^{-1}** for 2 TeV case

Muon polarization of 50% (?)

*) This corresponds to large initial muon flux at injection to booster of about **9.10^{13} μ/s**

Note: Extra power needed to run LHμC is not very large, **below 20 MW**

Antimuons are preferred for integration in LHC (but not quite for some physics)

Neutrino fluxes

LHμC will be very strong source of high energy neutrinos

Two aspects needs addressing/further studies:

- Protection against decays of 1 TeV muons along straight section of main muon ring requires detailed studies, but results for 1.5 TeV case of Muon Collider are very encouraging

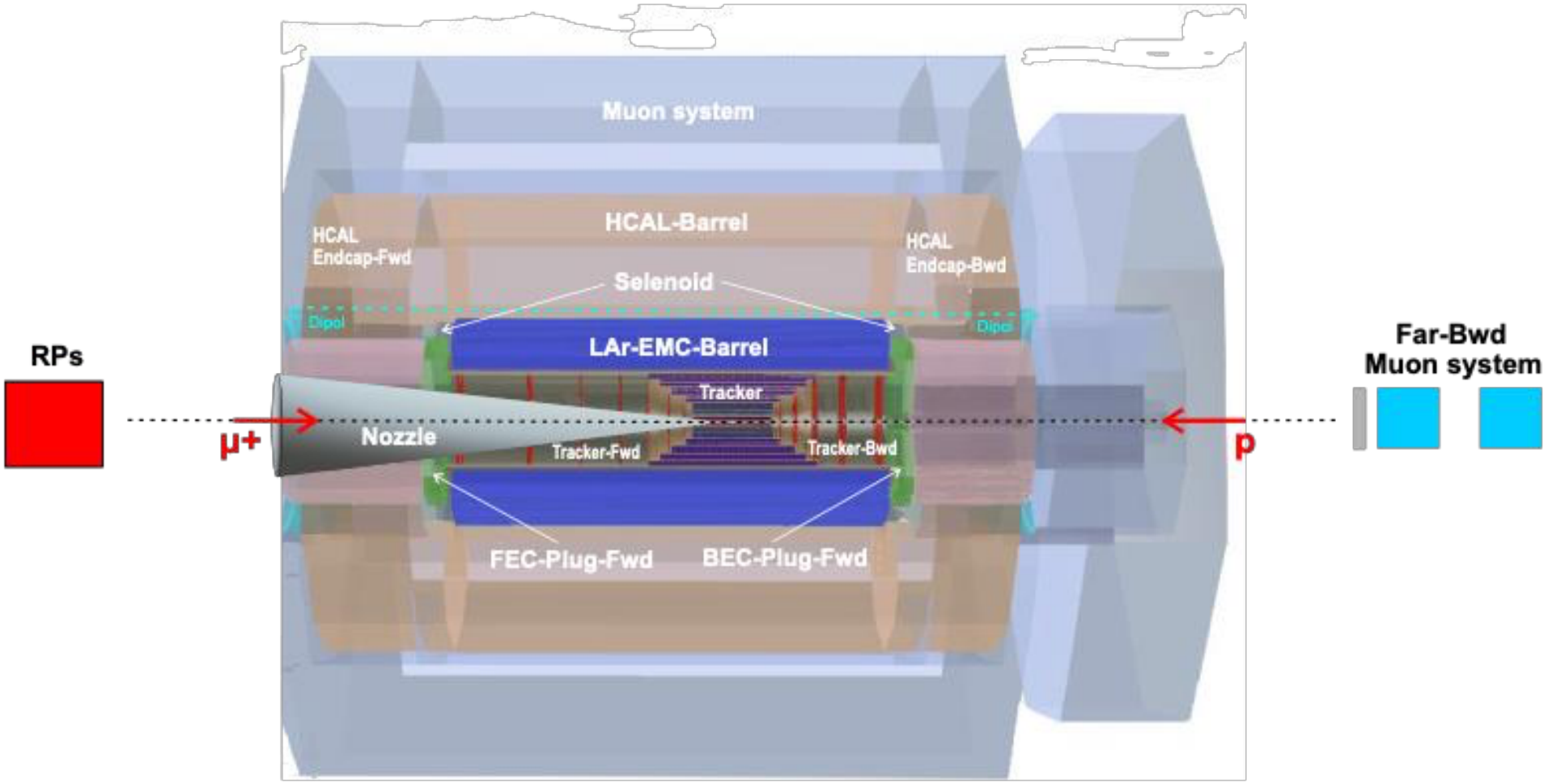
- **Exciting physics potential thanks to high fluxes of neutrinos** as result of μ^+ decays in Booster!

→ ν_e and $\bar{\nu}_\mu$ in wide energy range from sub-GeV up to hundreds of GeV:

$d\dot{N}/dE \simeq 10^{12}/E$ neutrinos of each type per second (per GeV of muon energy *E*)

Detector

Adapt LHeC design



Detector II

Highlights

- Physics motivated requirements: …
- Muon shielding: …
- Re-use of/synergy with LHeC detectors: …
- Far Backward Muon Detection: …

LHμC: Physics Highlights I

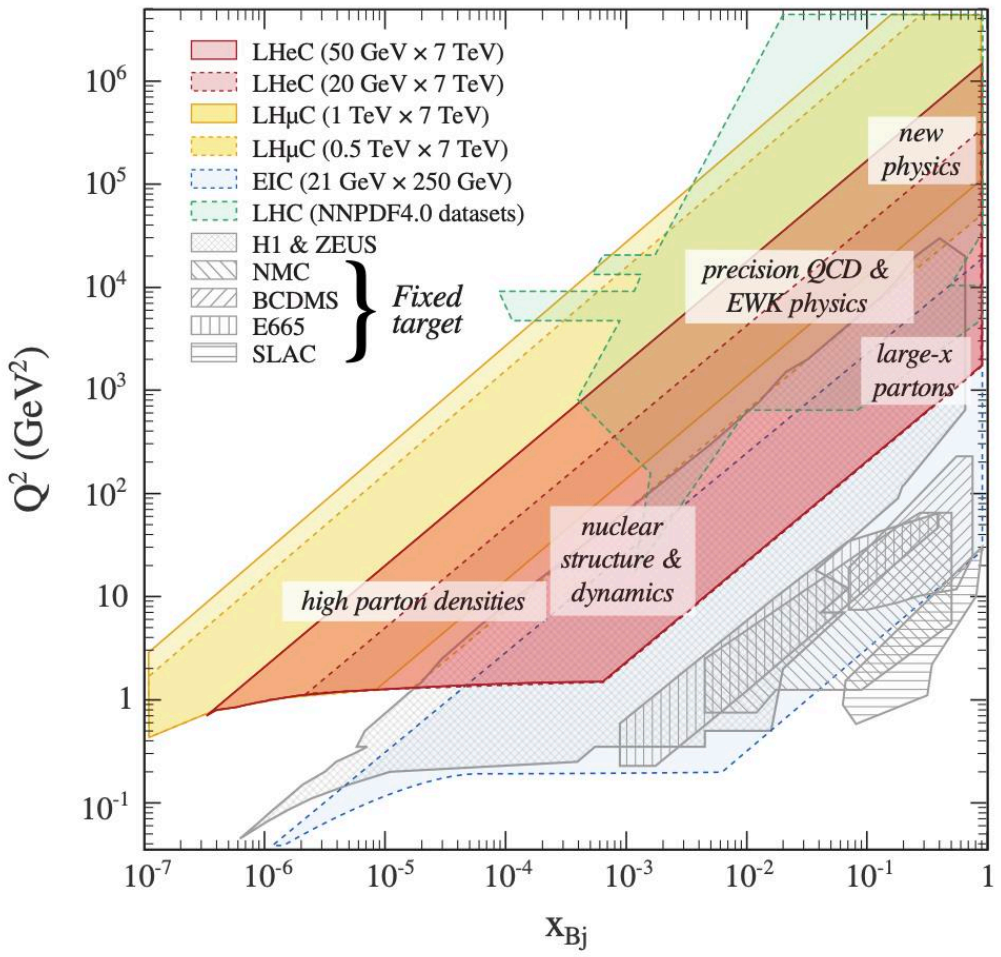


Figure 1. Available (x, Q^2) plane for the proposed LHeC and LHμC, in comparison to existing fixed target and collider datasets, and planned EIC kinematics reach. An inelasticity range of $4 \cdot 10^{-3} < y < 1$ is assumed to evaluate the lower and upper bounds.

Naturally, LHμC is **very powerful QCD machine:**

See talk by P. Kotko

It extends kinematical reach by order of magnitude wrt LHeC, **fully** entering domain of non-linear regime enabling unique and fundamental tests of QCD

Important inclusive measurements can be done **at low luminosity**

Unique precision studies (α_S, fully- & semi-exclusive production and more) – in clean experimental conditions as **event pileup is very low**!

Note: **muon-nucleus collisions** are also of large interest at LHμC

LHμC: Physics Highlights II

LHμC is not just powerful QCD machine, but thanks to its very high energy and luminosity it is also **powerful HET* Factory** (* Higgs-Electroweak-Top)

See talk by H. Khanpour

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(l\,p \to \nu_l\,H + \text{jet} + \text{X})$ [fb]	52	164	438	731
$\sigma(l\,p \to l\,H + \text{jet} + \text{X})$ [fb]	5.6	19	135	221
$\sigma(l\,p \to l\,t\bar{t} + \text{X})$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(l\,p \to \nu_l\,t + \text{X})$ [pb]	0.55	2.89	21.26	36.77

Table 1. Inclusive cross-sections for four major processes at the 20 and 50 GeV LHeC as well as 500 GeV and 1 TeV LHμC. In the LHeC cases an 80% electron beam polarisation is assumed

Note how much it is much superior to LH*e*C in top quark sector

LHμC: Physics Highlights III

Higgs case: “global impact” + hh

As summary: HELP4CERN

By end of 2026 *White Paper* will provide **comprehensive assessment of (very) recent developments** regarding lepton-hadron collisions at LHC, but with major two-fold goal in mind:

- Propose action plan for developing further LHμC concept/identify its main challenges/suggest next developments
- Foster world-wide collaboration/study group towards preparation **by 2028** of proper **"HELP4CERN Proposal"** in close collaboration with LH*e*C Study Group

Interested?

Contact: https://help4cern.web.cern.ch

HE *l*-*p*4CERN

Thank you for attention!